

AI water use vs animal products

Posted by u/the_elephant_stan • 0 points • 70 comments

Lately a lot of my carnist friends have latched onto the AI water use issue. This seems like an extension of their fake virtue signaling about water use around plant products like soy and almonds. The meat and dairy industry has done a great job of making people care about water use for everything *but* meat and dairy.

I want to be able to give them some reliable figures on meat and dairy water use. I see conflicting things. I have seen a figure for 1kg of beef equating to 15K liters of water, but then also seen things as low as several hundred. It's not always clear how much embodied water use these figures are taking into account.

Does anyone know where I can find reputable studies on this?

Comments

u/Dunkmaxxing • 3 points • 2026-03-21 02:09 UTC

Agriculture uses many magnitudes more water, many. It is not even remotely close at all. Anyone complaining about AI water use while supporting animal agriculture is a hypocrite beyond any defence.

u/Scarlet_Lycoris • 3 points • 2026-03-20 23:22 UTC

Tbh I don't think you have to argue either side. Why can't both be bad? I doubt this will shake your carnist friends to think over their diet if none of the other compelling reasons do. If you want to argue environmental impact of animal AG, you don't need to compare it to AI. It's also quite hard to measure what GPUs actually consume for inference.

[This](<https://bryantresearch.co.uk/insight-items/comparing-water-footprint-ai/>) seems to be putting AI water and Livestock water in perspective, though I can't fact check it rn. Might be worth a look though. They seem to link their sources at least.

u/the_elephant_stan • 2 points • 2026-03-21 01:20 UTC

Thank you for the link!

To be clear I'm not trying to argue that they aren't both bad, I'm trying to make the point that if you care about AI and not animal agriculture then you're full of shit about caring about water because the order of magnitude is so different.

They have decided that AI uses 1L of water per search. Doesn't matter if it's true or not. I'm trying to show them that it's dwarfed by animal agriculture.

u/[deleted] • 1 points • 2026-03-20 23:13 UTC

[removed]

u/AutoModerator • 1 points • 2026-03-20 23:13 UTC

Sorry, but [your comment]

(https://www.reddit.com/r/vegan/comments/1rz05e7/ai_water_use_vs_animal_products/obkr3yd/) has been removed for the following reason:

We are not accepting links to `x.com` at this time. Please find another source.

I am a bot, and this action was performed automatically. Please [contact the moderators of this subreddit] (/message/compose/?to=/r/vegan) if you have any questions or concerns.

u/OnYourToilet • 1 points • 2026-03-20 23:02 UTC

Getting numbers for the LLM side of this is going to be hard because the companies themselves are not very transparent about it. [Google](<https://cloud.google.com/blog/products/infrastructure/measuring-the-environmental-impact-of-ai-inference/>)did a [study](https://services.google.com/fh/files/misc/measuring_the_environmental_impact_of_delivering_ai_at_google_scale.pdf)they released but this is slightly misleading/ optimistic because they go by the median Gemini query, which might be very small/simple, it also doesn't look at the full 'life-cycle', so it doesn't take training or the making of the chips or the water used in power plants, etc. into account.

This [article by Mistral](<https://mistral.ai/news/our-contribution-to-a-global-environmental-standard-for-ai>) gives a very comprehensive overview, and I'd personally trust them more than Google, and they DO look at the full lifecycle but since they are based in France their CO2 estimates might also be optimistic.

But even if we go by the most pessimistic estimates, this is more in line of watching a short Youtube video, driving a few hundred meters or eating less than a tiny crumb of meat to consume as much water as an AI prompt.

Using the figure of [ourworldindata](<https://ourworldindata.org/grapher/scarcity-water-use-kcals>)given by many commenters here, 1000 kcal of beef require 12,273 L of water to produce, which means a 4 ounce burger requires about 3068 L.

If we go with Google's number of 0.26 mL for the median Gemini query and apply a factor 100 to it to get a more realistic (average) query, and another factor 2 to account for the training (the training consumes less than the inference when split over the millions of users, so this is a pessimistic estimate for the LLMs here), one query would consume about 52 mL of water (slightly higher than the Mistral estimate, so again, this is a pessimistic estimate as Google uses more efficient TPUs for their inference).

That means you'd need to make 59000 LLM queries just to consume as much water as a single 4 ounce beef burger. And again, I'm trying to be as biased against the LLM as possible here.